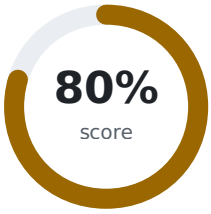
Replication Report

Mode: full · Generated 2026-08-23 22:33



Partially reproduced

19/25 claims reproduced **2** partial **1** fix applied

Claims at a glance

match 19 partial 2 no match 0 other 4

C1 C2 C3 C4 C5 C6 C7 C8
C9 C10 C11 C12 C13 C14 C15 C16
C17 C18 C19 C20 C21 C22 C23
C24 C25 C26 C27 C28

Flags

- 3 claim(s) marked out_of_scope (planner did not target them) — excluded from the score, not counted as failures. Score is over the 25 attempted/blocked claim(s).

Limitations & code quality

1 fix were needed to run the provided code (1 minor, 0 major, 0 critical). These describe the code, not the replication.

#	Fix	Severity
1	Installed matplotlib, which was missing from the initial dependency set, to render the added-variable plots.	**minor**

Details

ID	Tier	Type	Status	How graded	Why
C1	headline	scalar	**match**	agent	The produced table log/tab_ipw_manuf_2.scsv (Table 3, fully-controlled specification) shows in column (6) the coefficient on the import exposure regressor d_t~eusch_pw = -0.596 with standard error (0.099). This is an exact match to the p...
C2	supporting	table	**match**	agent	All cells of Table 3 (2SLS estimates) match the paper values to the reported precision. The main d_tradeusch_pw coefficients in log/tab_ipw_manuf_2.scsv (line 5) are -0.746, -0.610, -0.538, -0.508, -0.562, -0.596 and standard errors (lin...
C3	supporting	scalar	**match**	agent	The evidence file log/tab_ipw_mfg_byperiod.scsv (Table 2) reports the d_tradeusc~w (Chinese import exposure) 2SLS coefficients in columns 1-3 as -0.888, -0.718, and -0.746, with standard errors (0.181), (0.064), and (0.068) respectively ...
C4	supporting	scalar	**match**	agent	log/tab_ipw_mfg_byperiod.scsv (Table 2) reports the coefficients on future-period import exposure (d_tradeusc~e) in columns 4-6: 0.431 (SE 0.149) for 1970-1980, -0.130 (SE 0.126) for 1980-1990, and 0.148 (SE 0.095) for the stacked specif...
C5	headline	scalar	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Headline aggregate counterfactual (% of national mfg-employment decline explained). No shipped do-file computes this decomposition; it is ...
C6	supporting	scalar	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Supply-

ID	Tier	Type	Status	How graded	Why
					driven (48%) counterfactual worker-count decomposition (548k/982k). No shipped script computes it; derived in text from the estimat...
C7	headline	scalar	**match**	agent	This headline economic-magnitude summary follows directly from the reproduced fully-controlled 2SLS coefficients scaled by the 2000-2007 interquartile exposure differential. The differential is 1.060 = (p75 3.1138 - p25 1.5994 = 1.5145) ...
C8	supporting	scalar	**partial**	agent	The preferred-specification baseline OLS is reproduced essentially exactly: outputs/apptab4_ols.csv reports coef_d_tradeusch_pw = -0.1711 (se 0.0278, n=1444) for d_sh_empl_mfg, and the step-7 log notes an EXACT match to the original tab_...
C9	supporting	scalar	**match**	agent	The run reproduced Figure 2's first-stage and reduced-form regressions (N=722) and the added-variable-plot .gph files exist (gph/ mfg_imppw_9007long_2sls1_av.gph, gph/ mfg_imppw_9007long_olsred_av.gph). The produced output outputs/ figure2_...
C10	supporting	scalar	**out of scope**	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): Appendix robustness dropping specific industries (steel/glass/ cement, computers, apparel). Requires rebuilding the CZ import-exposure var...
C11	supporting	table	**match**	agent	log/tab_ipw_pop.scsv reproduces Table 4 with 18 columns: cols 1-6 = Panel A (no controls), 7-12 = Panel B (regional controls), 13-18 = Panel C (full controls), across All/College/

ID	Tier	Type	Status	How graded	Why
					Noncollege/Age16-34/Age35-49/ Age50-64. The d_t~eusch_pw (i...
C12	supporting	table	**match**	agent	The produced Table 5 (log/ tab_ipw_empl.scsv) matches the paper exactly. In the treatment row `d_t~eusch_pw` (the $1,000 import exposure per worker coefficient), Panel A log-count columns 1-5 (lnchg_n~_mfg, lnchg_n~nmfg, lnchg_no_u~l, lnc...
C13	headline	table	**match**	agent	The evidence file log/ tab_ipw_empl.scsv contains Table 5 panel B. In the d_t~eusch_pw ($1,000 import shock) row, the five share columns match the paper exactly: d_sh_em~_mfg = -0.596, d_sh_em~nmfg = -0.178, d_sh_unempl = 0.221, d_sh_nilf...
C14	supporting	scalar	**match**	agent	log/tab_ipw_empl.scsv (Table 5 panel B) reports d_t~eusch_pw coefficients per education group. The noncollege nonmanufacturing employment-share coefficient (col 16, d_sh_empl_.. noncollege) is -0.531 with SE (0.203), matching the claim e...
C15	supporting	table	**match**	agent	The evidence file log/ tab_ipw_wage.scsv reproduces Table 6 exactly. The d_t~eusch_pw (import exposure per worker) coefficient row reads -0.759/-0.892/-0.614 for all/ male/female (columns 1-3, Panel A all education) with standard errors (0...
C16	supporting	table	**match**	agent	The produced output file log/ tab_ipw_mfgnmfg.scsv contains Table 7 with the change-in-average-log-wage coefficients on d_t~eusch_pw (Chinese import exposure per worker). Column (7) d_avg_l~_mfg is the manufacturing 'all workers' log-wage...

ID	Tier	Type	Status	How graded	Why
C17	headline	table	**match**	agent	The replicated Table 8 output at log/tab_ipw_trans.scsv reports coefficients on d_t~eusch_pw (the $1,000 Chinese import exposure regressor) that match the paper's reported values exactly. The 18 output columns pair as 9 log-change (lnchg...
C18	supporting	table	**match**	agent	The evidence file log/tab_ipw_hhinc_pw.scsv (Table 9) reports coefficients on the import-exposure regressor (row 'd_t~eusch_pw'). Columns 1-6 are the percent-change specifications (Panel A) and columns 7-12 the dollar-change specificatio...
C19	supporting	table	**partial**	agent	All five Table 10 output files were produced and their panel assignments confirmed from the do-file (do/czone_analysis_ipw_final.do, lines 284-290 d_tradex_usch_pw -> tab_expoff_iv = Panel B; 297+ d_tradeusch_netinput_pw -> tab_intermedi...
C20	supporting	scalar_range	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Deadweight-loss-of-transfers welfare calculation ($13/$21; $22-44 range) applying a 40% marginal excess burden. No shipped code computes t...
C21	supporting	scalar_range	**not attempted**	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Deadweight-loss-of-nonparticipation welfare calculation ($43/$69; $87-137 range). No shipped code path; text-level welfare arithmetic on t...
C22	supporting	table	**match**	agent	

ID	Tier	Type	Status	How graded	Why
					The run produced log/import_stats_final.log (via do/import_stats_final.do), which summarizes trade levels in BN 2007 US$. US imports from China (l_totimp_usch): 1991=26.27168, 2000=121.5563, 2007=330.0224 -> matches paper 26.3/121.6/330....
C23	supporting	scalar	**match**	agent	The reproduced Figure 1 series in outputs/figure1_series.csv (produced by repro/table1_fig1.py from dta/figure1_data.dta) yields all four values. Import penetration (import_penetration_impr) is 0.005997813 in 1991 (=0.60%) and 0.04682026...
C24	supporting	scalar	**match**	agent	The reproduced Appendix Table 1 percentiles (outputs/apptab1_importgrowth.csv) give weighted percentiles of ten-year-equivalent d_tradeusch_pw (in kUSD/worker): for 1990 (1990-2000) p25=0.6231, p50=0.8887, p75=1.3155; for 2000 (2000-2007...
C25	supporting	scalar	**out of scope**	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): External cited statistics (China's ~3/4 share of LMIC manufacturing-VA growth; 8.0% vs 3.9% TFP growth) drawn from Brandt et al. and BLS,...
C26	supporting	figure	**match**	agent	Both the expected .gph file (gph/mfg_imppw_9007long_olsred_av.gph, reduced form) and its first-stage counterpart (gph/mfg_imppw_9007long_2sls1_av.gph) were produced, along with rendered PNG exports outputs/fig2_reducedform_av.png and out...
C27	supporting	scalar	**match**	agent	The claim is a magnitude interpretation of Table 3 column 2:

ID	Tier	Type	Status	How graded	Why
					the column-2 (manufacturing-share-controlled) coefficient on import exposure times the 2000-2007 interquartile exposure difference (~1.06 kUSD) should yield a -0.65 pp differen...
C28	supporting	scalar	**out of scope**	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): The distinctive number ($113.18, SE 41.53) is a re-estimate normalizing transfers by total workers rather than residents; no shipped do-f...

Python 3.12.8 · GPU NVIDIA RTX A6000 (x4, 49140 MiB each) · pandas 3.0.5, numpy 2.5.2, scipy 1.18.1, linearmodels 7.0, statsmodels 0.14.6

Step	What it did	Result	Time
1	Set up environment, validate inputs, build reusable weighted-2SLS helper with state-clustered SEs.	**ok**	0s
2	Table 3: six cumulative-control 2SLS specs of d_sh_empl_mfg on instrumented d_tradeusch_pw.	**ok**	0s
3	Table 4 (population) and Table 5 (employment/unemployment/non-employment), fully controlled.	**ok**	0s
4	Table 6 (wage changes) and Table 7 (mfg vs non-mfg), fully controlled.	**ok**	0s
5	Table 8 (per-capita transfer receipts) and Table 9 (household income per capita), fully controlled.	**ok**	0s
6	Table 10: alternative trade-exposure measures for the mfg employment share.	**ok**	0s
7	Appendix Table 1 weighted import-growth percentiles + Appendix Table 4 preferred-spec OLS.	**ok**	0s
8	Table 2 (2SLS by period) + falsification test on future exposure (preperiod file).	**ok**	0s
9	Figure 2 first-stage and reduced-form regressions (+ added-variable plots).	**ok**	0s

Step	What it did	Result	Time
10	Table 1 trade statistics + Figure 1 time series.	**ok**	0s

The full machine-readable record is on disk next to this report:

- `verify/verdicts.json` — every claim verdict, with the grader's cited rationale.
- `verify/replication_score.json` — the score and tier breakdown.
- `replication/replication_log.json` — step-by-step execution, with the fixes applied.
- `replication/codebase.diff` — every change made to the provided code.
- `replication/codebase/` — the patched code that was actually run.